

VAE-WSDE-Net: A Net-Load Forecasting Model for PV-ESS-Powered Game Data Centers via Manifold Learning

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Abstract: The exponential growth of digital entertainment and cloud gaming has made game data centers significant energy consumers, posing challenges to both operational costs and environmental sustainability. Integrating photovoltaic (PV) systems and battery energy storage systems (ESS) into data center microgrids is a promising solution. However, the inherent uncertainty of PV generation, coupled with fluctuating data center workloads, makes reliable power management difficult. Conventional forecasting methods often fail to provide reliable prediction intervals under complex scenarios like rapid workload changes or extreme weather. This limitation stems from their insufficient consideration of the underlying data physics. To address this, we propose the VAE-WSDE-Net model for net-load forecasting in PV-ESS-powered game data centers. First, a variational autoencoder (VAE) learns a meteorological-workload manifold from the data, establishing a foundation for physically consistent modeling. Then, we construct a WSDE-Net on this manifold. A key innovation is our tangential anisotropic noise injection mechanism, which aligns stochastic perturbations with the sensitivity of different physical features. This design provides more reliable prediction intervals. Additionally, we introduce a Wasserstein loss to resolve the gradient vanishing problem. Experimental results on real-world data demonstrate that our model achieves an average improvement of 38.1% in the Winkler Score over traditional methods, verifying its superior reliability and sharpness in net-load interval predictions.

Keywords: Uncertainty Prediction, Net-Load Forecasting, Game Data Center, Photovoltaic, Manifold Learning, VAE, SDE-Net

I. INTRODUCTION

The global energy transition is advancing rapidly, with renewable energy sources becoming central to achieving carbon neutrality. Concurrently, the digital economy, particularly the gaming and entertainment sector, is driving an exponential increase in data center energy demand [1]. Game data centers, which host cloud gaming platforms, online multiplayer servers, and streaming services, require

massive computational power, leading to a substantial and fluctuating carbon footprint. To mitigate this, integrating on-site Photovoltaic (PV) generation and Battery Energy Storage Systems (ESS) has emerged as a key strategy for creating sustainable, low-carbon data center microgrids.

However, the operation of such a "PV-ESS" powered data center faces significant challenges. The power supply from PV is inherently intermittent and uncertain due to its dependence on meteorological conditions. Simultaneously, the power demand (load) from the data center is highly volatile, driven by player activity, game updates, and time-of-day usage patterns. Effective microgrid energy management requires accurate forecasting of the *net-load*—the difference between data center load and PV generation. This net-load determines the required charging/discharging schedule for the ESS and the amount of grid power needed, directly impacting operational stability and cost. Reliable *probabilistic* forecasting, which provides prediction intervals (PIs) alongside point estimates, is crucial for risk-aware scheduling and ensuring uninterrupted gaming services.

Current uncertainty forecasting methods, such as Bayesian Neural Networks (BNN) [2], Monte Carlo Dropout [3], and Quantile Regression [4], often treat uncertainty as a statistical post-processing problem. Their estimates are largely "black-box" and lack physical interpretability. In our context, uncertainty stems not from random noise but from the physical randomness of meteorological dynamics and its interaction with the data center's workload physics. This decoupling from physical essence leads to unreliable PIs in out-of-distribution scenarios (e.g., a sudden thunderstorm combined with a new game launch). SDE-Net [5] models system evolution as a stochastic differential equation, offering a powerful framework. However, SDE-Net has limitations: 1. its isotropic noise injection overlooks complex feature correlations; 2. it doesn't fully leverage the data's underlying manifold structure.

To overcome these, we propose VAE-WSDE-Net, a net-load forecasting model for PV-ESS-powered game data centers. It innovatively integrates a Variational Autoencoder (VAE) with a Stochastic Differential Equation Network (SDE-Net). The VAE learns a low-dimensional latent manifold of the meteorological-workload data, disentangling physically coupled features. We then construct an SDE on this manifold, ensuring dynamic evolution is constrained

by the data's intrinsic geometry. Crucially, we propose a tangential anisotropic noise injection mechanism, proven to maintain physical consistency. Finally, Wasserstein loss is adopted to address gradient vanishing during training.

II. METHODOLOGY

A. SDE-Net

SDE-Net integrates Stochastic Differential Equations with neural networks for uncertainty quantification [5], [6]. The forward propagation is treated as a dynamical system described by an Itô SDE:

$$dx_t = f(x_t, \theta_f)dt + g(x_t, \theta_g)dW_t \quad (1)$$

where $f(\cdot)$ is the drift function (evolution trend), W_t is the Wiener process (noise source), and $g(\cdot)$ is the diffusion function (noise intensity). SDE-Net uses two neural networks for f and g . The training objective is to minimize prediction loss for in-distribution (ID) samples, minimize uncertainty for ID samples, and maximize uncertainty for out-of-distribution (OOD) samples [5]:

$$\begin{aligned} \min_{\theta_f} \mathbb{E}_{x_0 \sim P_{ID}} \mathbb{E}(\mathcal{L}(x_T)) + \min_{\theta_g} \mathbb{E}_{x_0 \sim P_{ID}} g(x_0; \theta_g) + \\ \max_{\theta_g} \mathbb{E}_{\tilde{x}_0 \sim P_{OOD}} g(\tilde{x}_0; \theta_g) \quad (2) \\ \text{s.t. } dx_t = f(x_t, \theta_f)dt + g(x_t, \theta_g)dW_t \end{aligned}$$

where $\mathcal{L}$ is the task loss.

B. Manifold Learning

The manifold hypothesis posits that high-dimensional data lies on a low-dimensional manifold $\mathcal{M}$ [7]. In our scenario, meteorological features (irradiance, temperature) and data center workload features (number of active users, CPU/GPU utilization per task) are not independent. They exhibit complex physical couplings in the original observation space. Deep generative models like the Variational Autoencoder (VAE) offer a powerful way to learn this intrinsic manifold [8], [9]. By integrating manifold learning with SDE-Net and using Wasserstein loss, our VAE-WSDE-Net ensures that stochastic evolution is constrained to the data's intrinsic manifold, aligning uncertainty with underlying physical principles.

III. VAE-WSDE-NET FOR GAME DATA CENTER NET-LOAD FORECASTING

The proposed VAE-WSDE-Net model integrates a VAE with an improved SDE-Net for net-load forecasting in PV-ESS-powered game data centers. The architecture is illustrated in Fig. 1. The input comprises historical meteorological data, historical data center load (e.g., user sessions, compute resource usage), and the state of charge (SoC) of the ESS. The output is the probabilistic forecast of the net-load for the next 24 hours.

A. Variational Autoencoder (VAE) for Feature Disentanglement

The VAE module maps high-dimensional input data $x \in \mathbb{R}^n$ (e.g., irradiance, temperature, number of active users, average session length) into a low-dimensional latent space $z \in \mathbb{R}^m$. The encoder $q_\phi(z|x)$ outputs parameters of a Gaussian distribution $(\mu, \log \sigma^2)$. The latent variable is sampled via the reparameterization trick: $z = \mu + \sigma \odot \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, I)$. The decoder $p_\theta(x|z)$ reconstructs the data.

The VAE is trained by optimizing the Evidence Lower Bound (ELBO):

1. Reconstruction Loss (MSE):

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_{q_\phi(z|x)} [\|x - \bar{x}\|^2] \quad (3)$$

2. KL Divergence:

$$\mathcal{L}_{\text{KL}} = -D_{\text{KL}}(q_\phi(z|x) \| p(z)) \quad (4)$$

The KL divergence encourages the latent space to be structured and dimensionally independent ($\text{Cov}(z) \approx I$), effectively disentangling the originally coupled physical variables. This provides an ideal foundation for subsequent stochastic modeling.

B. Improved SDE-Net on the Latent Manifold

The SDE-Net operates on the latent variable z_0 from the VAE. The drift network uses an LSTM to model the temporal dynamics of the net-load, while the diffusion network uses an FCN to quantify uncertainty.

1) Quantification of Epistemic and Aleatoric Uncertainty: Uncertainty is computed using Monte Carlo sampling. For each input, M forward passes through the SDE are performed, yielding samples $\{z_T^{(1)}, \dots, z_T^{(M)}\}$ and corresponding net-load predictions $\{y^{(1)}, \dots, y^{(M)}\}$. Epistemic and aleatoric uncertainties are estimated as [5]:

$$U_{\text{ep}} = \frac{1}{M-1} \sum_{m=1}^M (y^{(m)} - \bar{y})^2 \quad (5)$$

$$U_{\text{al}} = \mathbb{E}_{p(y_T|z_0, \theta_{f,g})} [\sigma(y_T)] \quad (6)$$

where $\bar{y}$ is the mean prediction, and $\sigma(y_T)$ is the estimated standard deviation from the diffusion network.

2) Tangential Anisotropic Noise Injection Mechanism:

The original SDE-Net's isotropic noise is physically unrealistic. We propose a tangential anisotropic mechanism. First, noise is constrained to the tangent space of the data manifold $T_z \mathcal{M}$. Second, a feature importance matrix $\mathbf{W}$ (derived from sensitivity analysis between latent factors and net-load) modulates noise intensity.

Proposition. Tangential Noise Injection Let $\mathcal{M} \subset \mathbb{R}^D$ be a smooth manifold of dimension $m < D$ representing the data's intrinsic structure. For a point $z \in \mathcal{M}$, the tangent space $T_z \mathcal{M}$ is an m -dimensional linear subspace of $\mathbb{R}^D$. The noise term in the SDE must satisfy $dW_t(z) \in T_z \mathcal{M}$ to ensure that the stochastic process

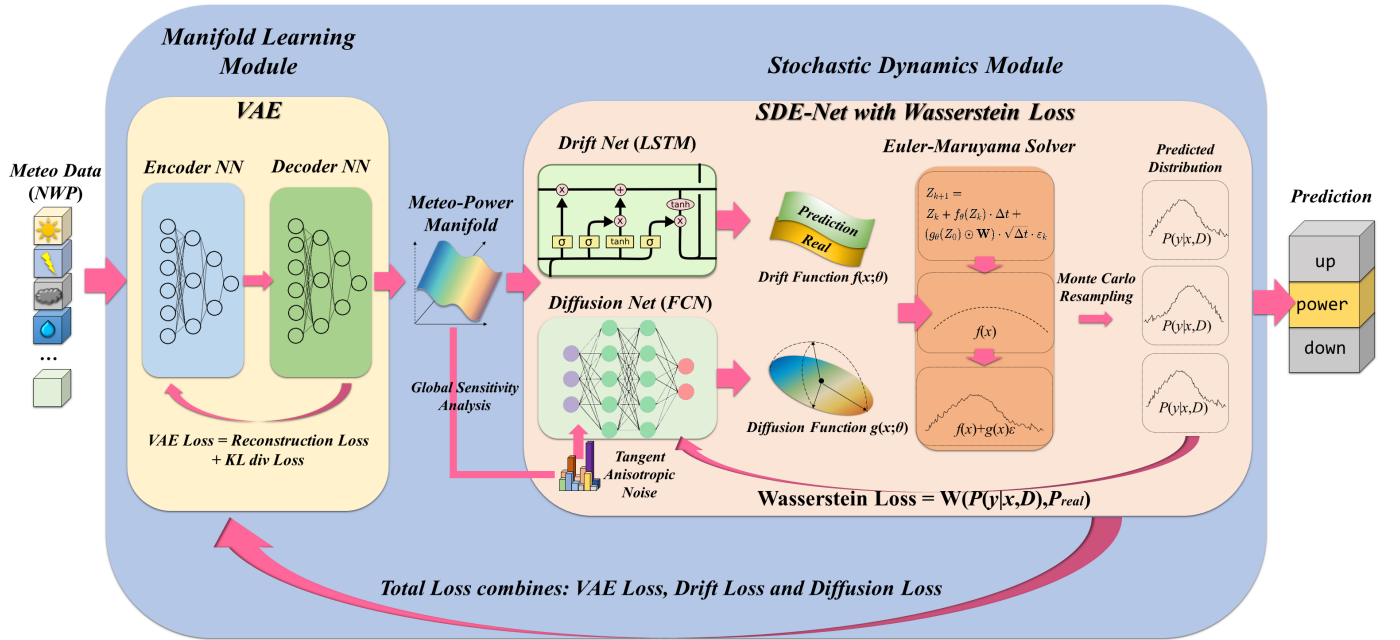


图 1. VAE-WSDE-Net for PV-ESS-Powered Game Data Centers. Input features (meteorological, workload, SoC) are mapped by the VAE onto a latent manifold $\mathcal{M}$. An LSTM drift network and an FCN diffusion network with tangential anisotropic noise operate on this manifold to produce probabilistic net-load forecasts.

z_t remains on the manifold $\mathcal{M}$ (i.e., the system state evolves in a physically consistent manner, not drifting into non-physical regions of the ambient space).

Proof. The VAE's decoder defines a smooth mapping from the latent space to the data space. The latent space itself is a Euclidean space where the VAE's regularization encourages a structured manifold. The tangent space at a point z can be approximated by the column space of the Jacobian of the decoder, $J_{p_\theta}(z)$. For the SDE evolution z_t , the infinitesimal change dz_t should lie in the span of the local coordinates of the manifold. By defining the noise term in the SDE as a linear combination of basis vectors of $T_z\mathcal{M}$ (e.g., through a projection operator $P_{T_z\mathcal{M}}$), we ensure that dz_t has no component normal to the manifold. More formally, one can consider the manifold as an embedded submanifold of $\mathbb{R}^m$ (the latent space). The standard Wiener process in $\mathbb{R}^m$ is already tangential. Our method ensures the anisotropic scaling via $\mathbf{W}$ and subsequent projection (implicitly enforced by operating on the VAE's structured latent space) keeps the dynamics confined to the learned manifold.

Based on this, the noise intensity is made feature-dependent:

$$\sigma_g = \gamma \cdot \text{softplus}(-\mathbf{W}z_0) \quad (7)$$

where σ_g is the noise intensity, γ is a scaling factor. This ensures weaker perturbations on important, deterministic latent factors (e.g., user count) and stronger perturbations on less important, secondary factors (e.g., specific cloud types).

3) **Objective Function:** The diffusion network's objective is to output low epistemic uncertainty for ID samples and high

uncertainty for OOD samples:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{z \sim P_{\text{ID}}} [\|g(z, \theta)\|_F^2] - \mathbb{E}_{z \sim P_{\text{OOD}}} [\|g(z, \theta)\|_F^2] \quad (8)$$

where $\|\cdot\|_F$ is the Frobenius norm.

4) **Euler-Maruyama Solver:** The SDE is solved numerically using the Euler-Maruyama method [10]. For the m -th sample path:

$$Z_{k+1}^{(m)} = Z_k^{(m)} + f(Z_k^{(m)}) \cdot \Delta t + (g(Z_k) \odot \mathbf{W}) \cdot \sqrt{\Delta t} \cdot \varepsilon_k^{(m)} \quad (9)$$

where $\varepsilon_k^{(m)}$ is standard Gaussian noise, and $\mathbf{W}$ is the feature importance matrix.

5) **Wasserstein Loss:** To replace the unstable negative log-likelihood (NLL) loss, we introduce the Wasserstein distance. For a batch of samples, a practical approximation is used [11], [12]:

$$\mathcal{L}_{\text{Wasserstein}} \approx \frac{1}{M} \sum_{m=1}^M (y - \hat{y}^{(m)})^2 \quad (10)$$

where y is the true net-load, and $\hat{y}^{(m)}$ is the prediction from the m -th forward pass. This loss provides smoother gradients and improves training stability.

C. Prediction Interval Construction

The $100(1-\alpha)\%$ prediction interval (PI) for net-load is derived from the empirical distribution of Monte Carlo samples:

$$\text{PI}_{\text{net-load}} = \left[Q_{\frac{\alpha}{2}} \left(\{y^{(m)}\}_{m=1}^M \right), Q_{1-\frac{\alpha}{2}} \left(\{y^{(m)}\}_{m=1}^M \right) \right] \quad (11)$$

where Q_p denotes the p -th sample quantile. This non-parametric approach ensures intervals reflect both types of uncertainty.

IV. EXPERIMENTS

A. Dataset & Environments

This study utilizes 577 days of data from a pilot green data center in China serving a major gaming company, from Jan 3, 2022, to July 31, 2023 (15-min resolution). The dataset includes:

- **PV Generation:** Measured power from on-site panels.
- **Data Center Load:** Aggregated power consumption of servers, cooling, and networking equipment.
- **ESS Status:** State of Charge (SoC) and charge/discharge rates.
- **Meteorological Data:** Historical and Numerical Weather Prediction (NWP) data.
- **Workload Metrics:** Number of active user sessions, average CPU/GPU utilization per session, and game launch schedules.

The target variable is the 24-hour ahead *net-load* (Data Center Load - PV Generation). The dataset was split chronologically into training (70%) and testing (30%). Experiments were run on an NVIDIA RTX 4090 GPU using PyTorch.

B. Feature Disentanglement and Manifold Discovery via VAE

The VAE disentangles intertwined meteorological and workload features. The result is shown in Fig. 2. The original observation space shows high entanglement, while the VAE's latent space reveals a structured, low-dimensional manifold with relatively independent latent variables.

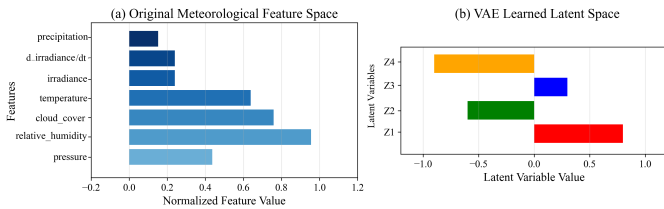


图 2. Feature Disentanglement of Meteorological and Workload Data via VAE.

C. Net-Load Uncertainty Prediction

1) **Metrics:** For point forecasts, MSE, MAE, and RMSE are used. For probabilistic forecasts, we use **Prediction Interval Normalized Average Width (PINAW)** for sharpness, **Prediction Interval Coverage Probability (PICP)** for reliability, and the **Winkler score** S_α as a comprehensive metric [13]:

$$S_\alpha = (U - L) + \frac{2}{\alpha} \max(L - y, 0, y - U) \quad (12)$$

where y is the true net-load, and U/L are the upper/lower bounds of the $100(1 - \alpha)\%$ PI. We set $\alpha = 0.10$ for 90% PIs.

2) **Results:** We compared VAE-WSDE-Net against BayesNN [2], QR-LSTM [14], MC Dropout (LSTM) [3], and SDE-Net [5]. All models forecasted 24-hour net-load. Results are in Table I, and Fig. 3 shows a sample forecast. VAE-WSDE-Net achieves the best balance

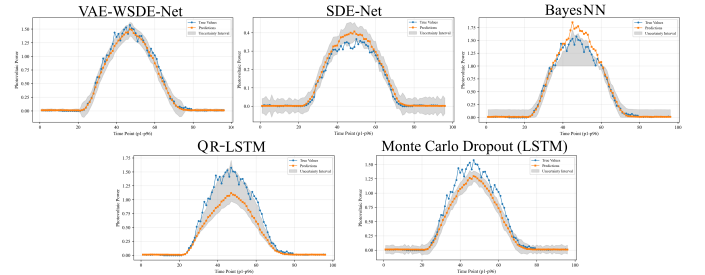


图 3. A representative example of 24-hour net-load prediction results.

between sharpness and coverage, reflected in its lowest Winkler Score. This confirms that manifold learning and physics-aware noise injection provide superior uncertainty quantification.

3) Case Study: A Game Launch on a Cloudy Day:

We analyzed a challenging day (User f6, July 5th, 2022) with fluctuating weather (intermittent rain, scattered clouds) coinciding with the launch of a major new game title, causing a rapid, sustained increase in data center workload. The performance of our model is benchmarked against the original SDE-Net, as shown in Fig. 4. SDE-Net's isotropic noise produces overly narrow and physically

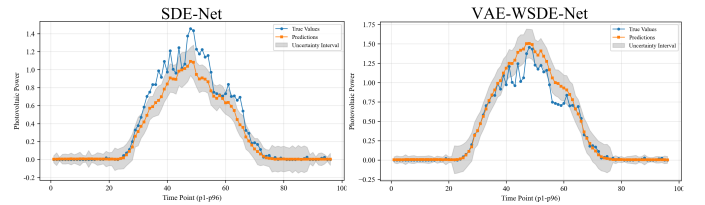


图 4. Comparative analysis of net-load prediction intervals during a game launch on a meteorologically volatile day.

inconsistent PIs during the sharp workload increase. In contrast, VAE-WSDE-Net, with its manifold-constrained dynamics, delivers sharper yet more responsive and physically plausible uncertainty estimates, effectively capturing the compound uncertainty from both weather and workload.

V. CONCLUSION

This paper proposed VAE-WSDE-Net, a novel manifold learning-based model for probabilistic net-load forecasting in PV-ESS-powered game data centers. By using a VAE to learn a physically meaningful latent manifold and injecting tangential anisotropic noise into an SDE-Net, our model ensures physically consistent uncertainty quantification. The introduction of Wasserstein loss further improves training stability.

表 I
COMPARISON BETWEEN VAE-WSDE-NET AND OTHER MODELS FOR NET-LOAD FORECASTING

Model	MSE	MAE	RMSE	PINAW	PICP	S_α
VAE-WSDE-Net	0.030	0.086	0.155	0.321	0.872	1.211
SDE-Net	0.046	0.104	0.190	0.183	0.697	2.889
BayesNN	0.050	0.107	0.200	0.141	0.657	3.677
QR-LSTM	0.050	0.110	0.200	0.164	0.252	1.681
MC-Dropout(LSTM)	0.047	0.107	0.196	0.424	0.851	1.800

Experiments on real-world data from a green data center demonstrate that VAE-WSDE-Net outperforms baseline methods in both point and interval forecast accuracy, achieving a superior balance between sharpness and reliability, especially under compound uncertainty from volatile weather and dynamic workloads. This work validates the critical role of physics-aware machine learning for sustainable and reliable operation of next-generation digital infrastructure.

Future work will focus on integrating reinforcement learning for optimal ESS scheduling based on these probabilistic forecasts and extending the model to federated learning settings across multiple data centers.

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